

Muthusamy Rajaguru

POSTDOCTORAL RESEARCHER

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Education

Postdoctoral Researcher

POSTDOCTORAL RESEARCHER IN IRENE VALENZUELA'S GROUP

IFT Madrid

2025-

Ph.D in Physics

DOCTORAL OF PHILOSOPHY IN PHYSICS COMPLETED UNDER TIMM WRASE'S SUPERVISION

Lehigh University

2020-2025

BS-MS in Physics

INTEGRATED BS-MS IN PHYSICS AT INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH THIRUVANANTHAPURAM

IISER TVM

2015-2020

All India Senior School Certificate Exam

CENTRAL BOARD OF SECONDARY EDUCATION

SBOA School and Junior College

2015

All India Secondary School Exam

CENTRAL BOARD OF SECONDARY EDUCATION

SBOA School and Junior College

2013

Publications

T-dualities and scale-separated AdS_3 in type I

ZHENG MIAO, MUTHUSAMY RAJAGURU, GEORGE TRINGAS, TIMM WRASE

- <https://arxiv.org/abs/2509.12801>

arXiv pre-print

2025

Characterisations of Scalar Potentials in String Theory Compactifications

DAVID ANDRIOT, MUTHUSAMY RAJAGURU, GEORGE TRINGAS

- <https://arxiv.org/abs/2501.17775> (JHEP 05 (2025) 046)

Journal of High Energy Physics

2025

Causality bounds from charged shockwaves in 5d

SERA CREMONINI, BRIAN McPEAK, MOHAMMAD MOEZZI, MUTHUSAMY RAJAGURU

- <https://arxiv.org/abs/2412.06891>

Journal of High Energy Physics

2024

Fully stabilized Minkowski vacua in the 2^6 Landau-Ginzburg model

MUTHUSAMY RAJAGURU, ANINDYA SENGUPTA, TIMM WRASE

- <https://arxiv.org/abs/2407.16756> (JHEP 10 (2024) 095)

Journal of High Energy Physics

2024

Moduli Stabilization in the 1^9 Model.

KATRIN BECKER, MUTHUSAMY RAJAGURU, ANINDYA SENGUPTA, JOHANNES WALCHER, TIMM WRASE

- <https://arxiv.org/abs/2406.03435> (JHEP 08 (2024) 069)

Journal of High Energy Physics

2024

On Asymptotic Dark Energy in String Theory

SERA CREMONINI, EDUARDO GONZALO, MUTHUSAMY RAJAGURU, YUEZHANG TANG, TIMM WRASE

- <https://arxiv.org/abs/2306.15714> (JHEP 09 (2023) 075)

Journal of High Energy Physics

2023

Automated Consistent Truncations and Stability of Flux Compactifications

DAVID ANDRIOT, PAUL MARCONNET, MUTHUSAMY RAJAGURU, TIMM WRASE

- <https://arxiv.org/abs/2209.08015> (JHEP 12 (2022) 026)

Journal of High Energy Physics

2022

Type IIB flux compactifications with $h^{1,1} = 0$

JACOB BARDZELL, EDUARDO GONZALO, MUTHUSAMY RAJAGURU, DANIELLE SMITH, TIMM WRASE

- <https://arxiv.org/abs/2203.15818> (JHEP 06 (2022) 166)

Journal of High Energy Physics

2022

Talks

Mining the LG FLUXscape

LOTUS & SWAMPLANDIA '26

- Presented a plenary talk at the Swampland workshop conducted in Catania, Italy.

Catania, IT

June, 2026

dS Critical Points and Rolling Solutions in Supergravity/String Theory

HARVARD SWAMPLAND SEMINAR

- Presented an invited talk in the Harvard Swampland seminar series.

Cambridge, MA

April, 2026

T-dualities and Scale Separated AdS_3 vacua in type I

ONLINE SWAMPLAND SEMINAR SERIES

- Presented an invited talk in the online Swampland seminar series.

Zoom

November, 2025

T-dualities and Scale Separated AdS_3 vacua

10TH IBS-IFT WORKSHOP

- Presented an invited talk in the 10th IBS-IFT Workshop conducted in IFT Madrid.

Madrid, Spain

November, 2025

Non-Geometric String Vacua

STRING PHENO '25

- Presented a talk in a parallel session during the String Pheno conference.

Boston, USA

July, 2025

Moduli stabilization and Landau-Ginzburg Models

HRI ALLAHABAD

- Presented a talk to the string group at HRI Allahabad.

Allahabad, India

August, 2024

Landau-Ginzburg models and the Swampland

IISER TVM

- Presented a talk in the physics seminar series at IISER TVM.

Trivandrum, India

August, 2024

Fully Stabilized Minkowski vacua in the 2^6 Landau-Ginzburg Model

THE LANDSCAPE VS THE SWAMPLAND

- Invited speaker at workshop on recent advances in the Landscape and the Swampland.

Vienna, Austria

July, 2024

Stabilizing Massless Fields in Landau- Ginzburg Models

STRING PHENO '24

- Presented a talk in a parallel session during the String Pheno conference.

Padova, Italy

June, 2024

Stabilizing Massless Fields in Landau- Ginzburg Models

SUSY' 24

- Presented a talk in the parallel session dedicated to String theory and Quantum field theory during the SUSY'24 conference.

Madrid, Spain

June, 2024

Non-Geometric String Compactifications and the Swampland

KU LEUVEN

- Presented a talk as part of the lunch seminar series organized by the theoretical physics group at KU Lueven.

Leuven, Belgium

June, 2024

Stabilizing Massless Fields in Landau-Ginzburg Models

SWAMPLANDIA '24

- Invited speaker at workshop on recent advances in the Swampland program.

Bavaria, Germany

May, 2024

Non-geoemtric String Compactifications and the Swampland

TEXAS A & M MITCHELL INSTITUTE SEMINAR

- Invited to present the Mitchell Institute Seminar by the high energy theory group at Texas A & M.

Texas A & M

January, 2024

Moduli stabilization in type IIB models with $h^{1,1} = 0$

STRING PHENO ONLINE SEMINARS

- Invited to present a talk in a bi-weekly seminar series conducted through the academic year.

Zoom

December, 2023

Moduli stabilization in the 1^9 Landau-Ginzburg Model

HARVARD SWAMPLAND VISITOR'S PROGRAM

- Invited to present a talk as part of the Harvard Swampland Visitors Program.

Harvard University

November, 2023

Moduli stabilization in type IIB models with $h^{1,1} = 0$

LAPTh ANNECY, HET SEMINAR

- Invited to present a seminar at the Laboratoire d'Annecy-le-Vieux de Physique Théorique.

LAPTh, Annecy

September, 2023

Non-geometric Flux Compactifications and the Swampland

STRING PHENOMENOLOGY ' 23

- Presented a talk based on upcoming work on moduli stabilization in a parallel session of the seminal conference in the field of String Phenomenology.

Institute of Basic Sciences, Daejeon

July, 2023

Dark Energy in String Theory

NEIGHBORHOOD WORKSHOP ON ASTROPHYSICS AND COSMOLOGY

- Presented a talk based on <https://arxiv.org/abs/2306.15714>.

Penn State, State College

April, 2023

Type IIB flux compactifications with $h^{1,1} = 0$ STRING PHENOMENOLOGY '22 • Presented a talk based on https://arxiv.org/abs/2203.15818 in a parallel session of the seminal conference in the field of String Phenomenology.	<i>Liverpool, UK</i> <i>July, 2022</i>
Probing for de Sitter Solutions in type IIB Supergravity WORKSHOP ON NON-LINEAR SUPERGRAVITY • Presented a talk based on https://arxiv.org/abs/2209.08015 .	<i>Technical University, Vienna</i> <i>January, 2022</i>

Fellowships and Awards

Dean’s Graduate Student Research Prize COLLEGE OF ARTS AND SCIENCES, LEHIGH UNIVERSITY • The Dean’s Graduate Student Research Prize is awarded once a year by the College of Arts and Sciences, Lehigh University.	<i>Lehigh University</i> <i>2024</i>
Lee Fellowship COLLEGE OF ARTS AND SCIENCES, LEHIGH UNIVERSITY • The Lee Fellowship is offered to meritorious physics and mathematics students to enable them to pursue research without teaching responsibilities.	<i>Lehigh University</i> <i>2022-present</i>
DTG-GO Travel Grant OFFICE OF INTERNATIONAL AFFAIRS, LEHIGH UNIVERSITY • Received a \$1000 travel grant from the Office of International Affairs to facilitate travel to various conferences and workshops in Europe.	<i>Lehigh University</i> <i>2024</i>
DTG-GO Travel Grant OFFICE OF INTERNATIONAL AFFAIRS, LEHIGH UNIVERSITY • Received a \$1500 travel grant from the Office of International Affairs to facilitate travel to Liverpool, UK for the String Phenomenology conference.	<i>Lehigh University</i> <i>2022</i>
Innovation in Science Pursuit for Inspired Research (INSPIRE) MINISTRY OF HUMAN RESOURCE AND DEVELOPMENT, INDIA • INSPIRE is a fellowship awarded by the Department of Science and Technology, India to meritorious students.	<i>IISER TVM</i> <i>2015-2020</i>

Schools and Conferences

Note, that the schools and conferences that appear in the section “Talks” are not repeated.

Strings and Cosmology DURATION : 3 DAYS • Attended workshop on advances in the string theory and cosmology.	<i>Annecy, France</i> <i>December, 2025</i>
Lotus and Swamplandia DURATION : 3 DAYS • Invited to attend workshop on advances in the Swampland program and string compactifications.	<i>Naxos, Greece</i> <i>June, 2025</i>
Cosmology Beyond ΛCDM DURATION : 3 DAYS • Attended workshop on advances in cosmology and early universe physics.	<i>Princeton, USA</i> <i>May, 2025</i>
Geometry and Phenomenology of String Compactifications DURATION : 1 DAY • Attended symposium focused on geometrical and phenomenological aspects of string compactifications	<i>New York City, USA</i> <i>December, 2024</i>
Amsterdam String Workshop DURATION : 2 WEEKS • Attended workshop on various advances in string theory and quantum gravity.	<i>Amsterdam, Netherlands</i> <i>June, 2024</i>
Surveying the Landscape DURATION : 3 DAYS • Invited to attend workshop on recent advances in the string landscape.	<i>UMass, Amherst</i> <i>April, 2024</i>
Swamplandia '23 DURATION : 4 DAYS • Early career scholarship awardee to attend workshop on recent advances in the swampland.	<i>IFT, Madrid</i> <i>September, 2023</i>

Advanced School on Quantum Field Theory and Quantum Gravity	ICISE, Quy Nhon
	DURATION : 3 WEEKS July, 2023
Attended a school which involved lectures on cutting edge research on flat space holography, cosmological correlators, swampland, AdS_3/CFT_2, black hole microstate counting, EFTs for condensed matter systems, amplitudes and machine learning.	
Theoretical Advanced Study Institute - TASI	University of Colorado, Boulder
	DURATION : 4 WEEKS June, 2023
Attended a school on the recent advances in generalized symmetries in field theory and the subject of topological field theory.	
String and Geometry	UPenn, Philadelphia
	DURATION : 3 DAYS March, 2023
Attended a workshop on recent advances in string compactifications and generalized symmetries.	
School on Superstring Theory and Related Topics	ICTP, Trieste
	DURATION : 1 WEEK April, 2021
Attended a school that involved lectures on holography, the swampland program and, anomalies.	
School on Astronomy, Gravitation and Cosmology	IIT Bombay
	DURATION : 4 WEEKS June, 2018
Attended a school which involved lectures on the interface between cosmology and particle physics, dark matter, general relativity and gravitational waves.	
National Science Camp, Vijyoshi	IISc Bangalore
	DURATION : 3 DAYS December, 2015
Attended an annual national science camp in India organised by KVPY in collaboration with INSPIRE program, IISc Bangalore and IISER-Kolkata for meritorious students.	

Extended Research Visits

Lehigh University	Lehigh University
	DURATION: 1 WEEK April, 2026
Invited to visit Lehigh University and present a seminar.	
Harvard Swampland Visitor’s Program	Harvard University
	DURATION: 1 WEEK April, 2026
Invited to visit the Swampland group at Harvard and present the Harvard Swampland Seminar.	
Visit to CERN	Geneva, Switzerland
	DURATION: 3 WEEKS November-December, 2025
Visited the theory group at CERN for research collaborations.	
Visit to IISER TVM	IISER TVM
	DURATION: 1 WEEK August, 2024
Visited the string theory group at IISER TVM and presented a talk and a lecture series.	
Visit to KU Leuven	KU Leuven
	DURATION: 1 WEEK June, 2024
Visited the thoeretical physics group at KU Leuven and presented a talk.	
Harvard Swampland Visitor’s Program	Harvard University
	DURATION: 3 WEEKS November, 2023
Invited to visit the Swampland group at Harvard and present a blackboard talk.	
Visitor to LAPTh, Annecy	LAPTh, Annecy
	DURATION: 1 WEEK September, 2023
Invited to visit Laboratoire d’Annecy-le-Vieux de Physique Théorique to present a seminar and for scientific collaboration.	

Referee Activity

JHEP	Referee for the Journal of High Energy Physics
EPJ C	Referee for the European Physical Journal - C

Skills

Programming C/C++, Python, ROOT, Matlab, Java

Tools Mathematica, LaTeX

Languages English, Tamil

References

Dr. Timm Wrase (Ph.D. Advisor)

Associate Professor of Physics

Lehigh University, USA

tiw419@lehigh.edu

Dr. Sera Cremonini

Professor of Physics

Lehigh University, USA

sec415@lehigh.edu

Dr. David Andriot

Senior Permanent Researcher (DR2)

LAPTh, CNRS, Annecy, France

andriot@lapth.cnrs.fr